

GATE IN

Instrumentation Engineering

TREND-BASED MOCK PAPER

— 01 —

Duration	3 Hours
Total Questions	65
Maximum Marks	100
Sections	8 subjects

Difficulty Distribution

Easy: 18 questions (28%)

Moderate: 39 questions (60%)

Difficult: 8 questions (12%)

GENERAL INSTRUCTIONS

GATE IN — Trend-Based Mock Paper

1. Examination Pattern

The paper contains 65 questions carrying a total of 100 marks. The duration of the examination is 3 hours.

2. Question Types

Questions are of three types:

- (a) Multiple Choice Questions (MCQ): One correct answer out of four options.
 - (b) Multiple Select Questions (MSQ): One or more correct answers out of four options. Each correct answer carries full marks. There is no negative marking.
 - (c) Numerical Answer Type (NAT): A real number answer to be entered. No options are provided.
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3. Marking Scheme

- 1-mark questions: +1 for correct, -0.33 for incorrect (MCQ only)
 - 2-mark questions: +2 for correct, -0.66 for incorrect (MCQ only)
 - 3-mark questions: +3 for correct, -0.66 for incorrect (MCQ only)
 - NAT questions carry no negative marking.
 - MSQ questions carry no negative marking.
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4. How to Answer

- For MCQ: Select exactly one option.
 - For MSQ: Select one or more options. Partial credit is NOT awarded.
 - For NAT: Enter the numerical answer in the provided space. Round off as specified.
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5. Rough Work

All rough work should be done only in the space provided in this question paper. Do not use additional sheets.

6. Calculator

An on-screen virtual calculator is available for use during the examination. Physical calculators are NOT permitted.

7. Important Notes

- Do not open the seal of the Question Booklet until you are instructed to do so.
 - The Question Booklet contains multiple pages. Verify the page count.
 - Immediately report any discrepancy in the Question Booklet to the invigilator.
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Q1. 1M [-0.33]

[General Aptitude]

20% of 200:

- (A) 20
- (B) 40
- (C) 80
- (D) 100

Q2. 1M [no -ve]

[Engineering Mathematics]

Determinant of 2x2 matrix:

Enter your answer here

Q3. 1M [-0.33]

[Sensors & Instrumentation]

LVDT measures:

- (A) Displacement
- (B) Velocity
- (C) Acceleration
- (D) Force

Q4. 1M [-0.33]

[Control Systems]

PID controller has:

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Q5. 1M [-0.33]

[Analog & Digital Electronics]

Instrumentation amp has:

- (A) 1 op-amp
- (B) 2
- (C) 3
- (D) 4

Q6. 1M [-0.33]

[Analog & Digital Electronics]

8-bit ADC has resolution:

- (A) 1/256
- (B) 1/128
- (C) 1/512
- (D) 1/64

Q7.

1M [no -ve]

[Communications & Process Control]

Closed-loop uses:

*Enter your answer here***Q8.**

1M [-0.33]

[Communications & Process Control]

FM uses:

- (A) Amplitude
- (B) Frequency
- (C) Phase
- (D) None

Q9.

1M [-0.33]

[Measurement Systems]

Moving coil instrument measures:

- (A) AC only
- (B) DC only
- (C) Both
- (D) None

Q10.

1M [-0.33]

[Control Systems]

PID controller has:

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Q11.

1M [-0.33]

[Sensors & Instrumentation]

LVDT measures:

- (A) Displacement
- (B) Velocity
- (C) Acceleration
- (D) Force

Q12.

1M [-0.33]

[General Aptitude]

20% of 200:

- (A) 20
- (B) 40
- (C) 80
- (D) 100

Q13.

1M [-0.33]

[Control Systems]

PID controller has:

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Q14.

1M [no -ve]

[Engineering Mathematics]

Determinant of 2x2 matrix:

*Enter your answer here***Q15.**

1M [-0.33]

[Sensors & Instrumentation]

LVDT measures:

- (A) Displacement
- (B) Velocity
- (C) Acceleration
- (D) Force

Q16.

1M [-0.33]

[Analog & Digital Electronics]

Instrumentation amp has:

- (A) 1 op-amp
- (B) 2
- (C) 3
- (D) 4

Q17.

1M [-0.33]

[Sensors & Instrumentation]

LVDT measures:

- (A) Displacement
- (B) Velocity
- (C) Acceleration
- (D) Force

Q18.

1M [-0.33]

[Engineering Mathematics]

Determinant of 2x2 matrix:

- (A) $ad-bc$
- (B) $ad+bc$
- (C) $a+d$
- (D) ab

Q19.

1M [-0.33]

[Sensors & Instrumentation]

LVDT measures:

- (A) Displacement
- (B) Velocity
- (C) Acceleration
- (D) Force

Q20.

1M [-0.33]

[Engineering Mathematics]

Determinant of 2x2 matrix:

- (A) $ad-bc$
- (B) $ad+bc$
- (C) $a+d$
- (D) ab

Q21.

1M [-0.33]

[General Aptitude]

20% of 200:

- (A) 20
- (B) 40
- (C) 80
- (D) 100

Q22.

1M [-0.33]

[Communications & Process Control]

FM uses:

- (A) Amplitude
- (B) Frequency
- (C) Phase
- (D) None

Q23.

1M [-0.33]

[General Aptitude]

20% of 200:

- (A) 20
- (B) 40
- (C) 80
- (D) 100

Q24.

1M [-0.33]

[Sensors & Instrumentation]

LVDT measures:

- (A) Displacement
- (B) Velocity
- (C) Acceleration
- (D) Force

Q25.

1M [no -ve]

[Analog & Digital Electronics]

8-bit ADC has resolution:

*Enter your answer here***Q26.**

1M [-0.33]

[Analog & Digital Electronics]

Instrumentation amp has:

- (A) 1 op-amp
- (B) 2
- (C) 3
- (D) 4

Q27.

1M [-0.33]

[Control Systems]

PID controller has:

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Q28.

1M [-0.33]

[Signals and Systems]

FFT computes:

- (A) DFT
- (B) DTFT
- (C) CTFT
- (D) Laplace

Q29.

1M [-0.33]

[Analog & Digital Electronics]

Instrumentation amp has:

- (A) 1 op-amp
- (B) 2
- (C) 3
- (D) 4

Q30.

1M [-0.33]

[Engineering Mathematics]

Determinant of 2x2 matrix:

- (A) $ad-bc$
- (B) $ad+bc$
- (C) $a+d$
- (D) ab

Q31.

2M [-0.66]

[General Aptitude]

20% of 200:

- (A) 20
- (B) 40
- (C) 80
- (D) 100

Q32.

2M [-0.66]

[Communications & Process Control]

Closed-loop uses:

- (A) No feedback
- (B) Feedback
- (C) Feedforward
- (D) None

Q33.

2M [no -ve]

[Communications & Process Control]

Closed-loop uses:

*Enter your answer here***Q34.**

2M [-0.66]

[Measurement Systems]

Moving coil instrument measures:

- (A) AC only
- (B) DC only
- (C) Both
- (D) None

Q35.

2M [-0.66]

[Communications & Process Control]

FM uses:

- (A) Amplitude
- (B) Frequency
- (C) Phase
- (D) None

Q36.

2M [-0.66]

[Control Systems]

PID controller has:

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Q37.

2M [no -ve]

[Engineering Mathematics]

Determinant of 2x2 matrix:

*Enter your answer here***Q38.**

2M [-0.66]

[Sensors & Instrumentation]

LVDT measures:

- (A) Displacement
- (B) Velocity
- (C) Acceleration
- (D) Force

Q39.

2M [-0.66]

[Analog & Digital Electronics]

Instrumentation amp has:

- (A) 1 op-amp
- (B) 2
- (C) 3
- (D) 4

Q40.

2M [no -ve]

[Analog & Digital Electronics]

8-bit ADC has resolution:

*Enter your answer here***Q41.**

2M [no -ve]

[Signals and Systems]

FFT computes:

- (A) DFT
- (B) DTFT
- (C) CTFT
- (D) Laplace

Q42.

2M [-0.66]

[Sensors & Instrumentation]

LVDT measures:

- (A) Displacement
- (B) Velocity
- (C) Acceleration
- (D) Force

Q43.

2M [-0.66]

[Measurement Systems]

Moving coil instrument measures:

- (A) AC only
- (B) DC only
- (C) Both
- (D) None

Q44.

2M [-0.66]

[Communications & Process Control]

Closed-loop uses:

- (A) No feedback
- (B) Feedback
- (C) Feedforward
- (D) None

Q45.

2M [no -ve]

[Communications & Process Control]

FM uses:

- (A) Amplitude
- (B) Frequency
- (C) Phase
- (D) None

Q46.

2M [no -ve]

[Sensors & Instrumentation]

LVDT measures:

*Enter your answer here***Q47.**

2M [-0.66]

[Control Systems]

PID controller has:

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Q48.

2M [-0.66]

[Control Systems]

PID controller has:

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Q49.

2M [-0.66]

[Analog & Digital Electronics]

Instrumentation amp has:

- (A) 1 op-amp
- (B) 2
- (C) 3
- (D) 4

Q50.

2M [no -ve]

[Analog & Digital Electronics]

8-bit ADC has resolution:

*Enter your answer here***Q51.**

2M [no -ve]

[Communications & Process Control]

FM uses:

- (A) Amplitude
- (B) Frequency
- (C) Phase
- (D) None

Q52.

2M [-0.66]

[Analog & Digital Electronics]

Instrumentation amp has:

- (A) 1 op-amp
- (B) 2
- (C) 3
- (D) 4

Q53.

2M [no -ve]

[Engineering Mathematics]

Determinant of 2x2 matrix:

*Enter your answer here***Q54.**

2M [-0.66]

[Sensors & Instrumentation]

LVDT measures:

- (A) Displacement
- (B) Velocity
- (C) Acceleration
- (D) Force

Q55.

2M [no -ve]

[Signals and Systems]

FFT computes:

- (A) DFT
- (B) DTFT
- (C) CTFT
- (D) Laplace

Q56.

2M [no -ve]

[Control Systems]

PID controller has:

Enter your answer here

Q57.

2M [-0.66]

[Communications & Process Control]

FM uses:

- (A) Amplitude
- (B) Frequency
- (C) Phase
- (D) None

Q58.

2M [-0.66]

[Signals and Systems]

FFT computes:

- (A) DFT
- (B) DTFT
- (C) CTFT
- (D) Laplace

Q59.

2M [-0.66]

[Control Systems]

PID controller has:

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Q60.

2M [-0.66]

[General Aptitude]

20% of 200:

- (A) 20
- (B) 40
- (C) 80
- (D) 100

Q61.

2M [-0.66]

[Communications & Process Control]

Closed-loop uses:

- (A) No feedback
- (B) Feedback
- (C) Feedforward
- (D) None

Q62.

2M [no -ve]

[Control Systems]

PID controller has:

- (A) 1
- (B) 2
- (C) 3
- (D) 4

Q63.

2M [-0.66]

[Engineering Mathematics]

Determinant of 2x2 matrix:

- (A) $ad-bc$
- (B) $ad+bc$
- (C) $a+d$
- (D) ab

Q64.

2M [-0.66]

[Communications & Process Control]

FM uses:

- (A) Amplitude
- (B) Frequency
- (C) Phase
- (D) None

Q65.

2M [-0.66]

[Control Systems]

PID controller has:

- (A) 1
- (B) 2
- (C) 3
- (D) 4